

Response to the Editors and Reviewers

IEEE Transactions on Circuits and Systems for Video Technology

Manuscript TCSVT-29693-2026: *Beyond Homogeneous Adaptation: Decoupled Control for Sample- and Layer-Aware Test-Time Robustness in Temporally Correlated Streams*

Dear Editor-in-Chief, Senior Area Editor, Associate Editor, and Reviewers,

Thank you for your careful evaluation and constructive suggestions. We appreciate the positive assessment of the motivation, technical design, and long-horizon evidence of the Decoupled Control Framework (DCF). Your comments helped us substantially strengthen the analytical interpretation, experimental validation, computational cost assessment, positioning, and reproducibility of DCF.

We summarize the main revisions in three groups below, then respond to each comment and identify the corresponding manuscript changes.

1. Probe interpretation and layer behavior. We strengthened the local directional-sensitivity interpretation of PCS through analytical expansion and introduced controlled shape–texture and Colored-MNIST shortcut experiments, matched-coverage routing diagnostics, and frequency/strength sensitivity analyses (Tables IV–VI, Figs. 5 and 12(c)). These additions establish PCS as a structured directional-sensitivity signal and demonstrate that probe responsiveness provides shortcut-sensitive routing information beyond predictive entropy alone. We also incorporated retention-gate heatmaps over time and corruption types, evaluated control granularity across network depth, and benchmarked our gradient-square proxy against Expected Fisher and a Hutchinson-based second-order approximation (Fig. 9, Table IX).

2. Geometry repair and computational cost. We compared RGR against moment matching, prototype contrastive alignment, and uniform-prior optimal transport, and characterized its behavior across routing ratios from 10% to 90%, demonstrating stability even when routed samples are sparse or dominant (Tables VII–VIII). Furthermore, we benchmarked FLOPs, peak memory, and latency alongside representative baselines. On an NVIDIA A100, the lightweight variant DCF-Lite reduces FLOPs by 61.8% and peak memory by 58.6% while achieving 42.06% accuracy (only 1.42 points below full DCF and 3.16 points above SAR), establishing a practical path toward lower-cost online adaptation (Table XI).

3. Empirical rigor, positioning, and reproducibility. To rigorously evaluate generalizable representation preservation, we added DomainNet-126 as a distinct multi-domain benchmark alongside ImageNet-C. Across five matched runs, paired t -tests confirm that DCF achieves statistically significant aggregate cross-domain transfer gains on both ImageNet-C ($p = 1.39 \times 10^{-4}$) and DomainNet-126 ($p = 2.75 \times 10^{-4}$) (Fig. 11). In direct response to editorial guidance, we explicitly discuss four closely related TCSVT studies ([29], [30], [31], [36]) and clarify the distinctive coupled sample–layer control formulation of DCF, while incorporating the suggested sensor-stream literature ([19]–[21]) and recent continual-TTA advances ([28], [38]). Finally, we provide full technical clarifications regarding trainable parameters, CLR block grouping, routing regions, threshold selection, and Sinkhorn OT settings.

Below, we respond to each editorial request and every reviewer comment in detail. We hope that the revised manuscript and our detailed responses satisfactorily address all concerns, and we respectfully submit the revised manuscript for your further consideration.

Online revision review: tcsvt-29693-2026.xhy.im. Each response includes direct links to its online reply and the corresponding PDF comparison, with additional figures, tables, and mapped manuscript evidence.

Sincerely,

Yuanyuan Pu

Corresponding author, on behalf of all authors

Roadmap of Responses to Editors and Reviewers

Editor / Reviewer Group	Direct Jump
Editor-in-Chief (EIC)	Jump → p. ??
Senior Area Editor (SAE)	Jump → p. ??
Associate Editor (AE)	Jump → p. ??
Reviewer 1 (R1)	Jump → p. ??
Reviewer 2 (R2)	Jump → p. ??

Response to the Editor-in-Chief

Related TCSVT studies and the distinctive contribution of DCF

Before your new submission, you have to answer two questions clearly in your revised manuscript and responses: a) what are the 3-5 papers published in the IEEE Transactions on Circuits and Systems for Video Technology, which are most closely related to your manuscript; b) what is distinctive / new about your current manuscript related to these previously published papers.

► [Online response \(EIC\)](#) | ► [PDF comparison](#)

► Response:

We sincerely appreciate the Editor-in-Chief’s constructive guidance. Following this guidance, we have explicitly identified the closely related IEEE TCSVT studies and clarified how DCF differs from and advances beyond these works in the revised manuscript. This comparison also helps position the contribution of DCF more clearly within the scope of IEEE TCSVT.

Below, we directly answer the two questions.

1. Response to Question a): Closely Related Papers Published in IEEE TCSVT

In the revised Section II (“Related Work”), we have explicitly incorporated and analyzed four closely related studies recently published in IEEE TCSVT (spanning 2024 to 2026) that address test-time adaptation, target-evidence reliability, online model adaptation, or target-side representation refinement:

- 1) **QED** (Liu *et al.*, IEEE TCSVT 2024) [31]: *Question Type-Aware Debiasing for Test-Time Visual Question Answering Model Adaptation*.
Focus & Mechanism: QED addresses test-time bias in VQA by leveraging question-type-aware entropy and negative samples to identify predictions likely to be influenced by language priors. Its primary control determines whether a test instance should contribute to adaptation according to task-specific question–answer structure.
- 2) **MetaBN** (Liu and Yang, IEEE TCSVT 2025) [29]: *Test-Time Adaptation for Real-World Video Adverse Weather Restoration With Meta Batch Normalization*.
Focus & Mechanism: MetaBN develops adaptation capability through source-stage meta-learning of normalization affine parameters and applies self-supervised updates online for adverse-weather video restoration. Its adaptation mechanism is therefore closely tied to source-stage preparation of normalization behavior.
- 3) **CMDA** (Zhang *et al.*, IEEE TCSVT 2026) [30]: *Collaborative Model and Data Adaptation at Test Time*.
Focus & Mechanism: CMDA couples predictive model adaptation with diffusion-based data refinement in a bidirectional loop: current predictions guide reverse-diffusion sample generation, and the generated samples subsequently refine the predictive model.
- 4) **MS-TTA** (Han *et al.*, IEEE TCSVT 2026) [36]: *Unleashing the Potential of All Test Samples: Mean-Shift Guided Test-Time Adaptation*.
Focus & Mechanism: MS-TTA exploits information from test samples without conventional online parameter optimization by applying mean-shift refinement to CLIP embeddings and maintaining a feature cache during inference.

2. Response to Question b): Distinctive Novelty and Advances of DCF

DCF’s distinctive contribution is to formulate long-horizon test-time adaptation under temporally correlated streams as a coupled sample–layer feedback control problem, and to operationalize this formulation through a decoupled route–adapt–retain architecture.

Rather than treating unreliable target evidence and parameter drift as two independent failure sources, DCF addresses their interaction as an error-amplifying feedback process: the current target evidence determines the adaptation update, the retained update reshapes subsequent predictions and sample selection, and its effect accumulates heterogeneously across network depth. This formulation motivates three coordinated control decisions within the same online loop: *which evidence should drive direct adaptation, how non-trusted evidence should still contribute to adaptation, and where the resulting candidate update should be allowed to persist.*

None of the four closely related TCSVT studies above jointly addresses these three decisions within one recurrent online adaptation process:

- **From Sample Selection to Heterogeneous Evidence Routing.** QED regulates adaptation evidence according to VQA-specific question–answer structure, while MetaBN develops test-time adaptability through source-stage normalization meta-learning. DCF instead addresses temporally correlated recognition streams in which confidence alone may not provide sufficient evidence reliability. **PSR** jointly evaluates predictive entropy and responsiveness under structured Fourier stress, distinguishing confident, stress-responsive samples from confident but stress-inert candidates. The contribution is therefore not merely an additional filtering criterion, but an explicit routing decision over heterogeneous target evidence.
- **From Discarding Non-Trusted Samples to Alternative Supervision.** DCF further distinguishes between evidence that is unsuitable for direct pseudo-label supervision and evidence that is entirely uninformative for adaptation. **RGR** fundamentally changes the role of routed-away samples: instead of discarding them or forcing them into direct pseudo-label supervision, it converts them into soft class-geometry constraints through entropic optimal transport. This explicitly separates “not trusted for direct supervision” from “not useful for adaptation.” In contrast, CMDA establishes a model–generator refinement loop through diffusion-based data adaptation, whereas MS-TTA refines and caches test representations without conventional online parameter optimization.
- **From Parameter Updating to Layer-Dependent Update Persistence.** DCF also separates the generation of an adaptation update from the decision of whether that update should persist. **CLR** treats every sample-side optimization result only as a candidate update and selectively determines where that update is retained across network depth using current source–candidate curvature mismatch. This suppresses error-amplifying drift in source-sensitive layers while preserving useful plasticity elsewhere, rather than applying a single global stabilization rule to the entire model.

Importantly, the distinction of DCF does not arise from any of these components in isolation. It arises from coordinating evidence routing, alternative use of non-trusted samples, and layer-dependent update persistence within the same recurrent adaptation loop. In this sense, DCF moves beyond homogeneous adaptation control: it controls not only whether the model adapts, but *what evidence drives adaptation, what role the remaining evidence plays, and where the resulting update is allowed to accumulate*. This coupled sample–layer control is the principal distinction of DCF from the closely related TCSVT studies above.

Evidence for the Coordinated Design. The revised experiments evaluate these controls both individually and as a coupled system. Table III isolates the contributions of PSR, RGR, and CLR. Tables IV–VI analyze Fourier-probe responsiveness, shortcut sensitivity, and selected-sample reliability. Tables VII–VIII compare geometry-repair alternatives and characterize routing ratios from 10% to 90%. Fig. 9 and Table IX expose layer-wise retention behavior and compare alternative curvature proxies. Table II evaluates stability over the 225-domain long-horizon stream, while Fig. 11 examines post-adaptation transfer to unseen domains. Table XI further reports computational cost and the DCF-Lite approximation, which reduces FLOPs by 61.8% and peak memory by 58.6% relative to full DCF while retaining strong long-horizon accuracy. Taken together, these results support the central contribution of DCF as a coordinated sample–layer control framework rather than a collection of independent filtering, alignment, or regularization components.

Changes in the manuscript: Section I (p. 2); Section II (p. 3); Section III-B (p. 3, Eqs. (2)–(3)); Section IV-C (pp. 7–12, Tables II–IX, XI; Figs. 9, 11, 12); References [29], [30], [31], [36] (p. 13).

Selected Revised Text in Section II (Page 3):

“Related studies further explore question-conditioned debiasing [31], meta-trained normalization for adverse-weather adaptation [29], diffusion-based collaborative model–data adaptation [30], and training-free mean-shift refinement of test representations [36]. In contrast, DCF controls the coupled evolution of target evidence and model parameters in temporally correlated recognition streams. PSR routes evidence by entropy and Fourier stress responsiveness; RGR converts routed-away samples into soft OT geometry constraints; and CLR determines where the resulting candidate update persists.”

“DCF formulates online adaptation as a coupled sample–layer feedback process, jointly controlling which evidence drives direct supervision, how routed-away samples constrain target geometry, and where candidate updates persist. This route–adapt–retain design interrupts the error-amplifying feedback that causes long-horizon collapse.”

Response to the Senior Area Editor

Relevance, completeness, and ablation depth

The paper proposes the use of a decoupled control system for reducing bias in target evidence in test-time adaptation using online methods for temporally correlated image streams. The topic is relevant to TCSVT. The paper is complete with contributions, experiments, results, and a limited ablation study.

► [Online response \(SAE\)](#) | ► [PDF comparison](#)

► Response:

Thank you for recognizing the relevance and completeness of the work. We expanded the ablation study in Section IV-C with controlled PSR diagnostics; RGR alternatives, prior ablation, and routing-ratio stress tests; CLR gate visualizations, granularity and proxy comparisons; cross-domain statistics; and DCF-Lite efficiency analysis. These additions establish the contribution of each control mechanism, reveal their interactions, and quantify the accuracy–efficiency trade-off.

Changes in the manuscript: Section IV-C, Tables IV–IX and XI, pp. 8–12; Figs. 9, 11, and 12, pp. 10–12.

Response to the Associate Editor

Thank you for identifying the analyses needed for this major revision. We address the five substantive concerns below; the corresponding Reviewer 1 responses provide the detailed evidence.

Associate Editor: Comment Overview and Navigation

Item	Comment Title	Direct Jump
AE1	Justification and sensitivity of the Fourier probe	AE1 (p. ??) → [R1.1]
AE2	Alternatives and failure cases for geometry repair	AE2 (p. ??) → [R1.2]
AE3	Computational cost and practical approximations	AE3 (p. ??) → [R1.3]
AE4	Behavior of the layer retention gates	AE4 (p. ??) → [R1.4]
AE5	Statistical evidence for cross-domain transfer	AE5 (p. ??) → [R1.6]

AE1: Justification and sensitivity of the Fourier probe

The Fourier-based stress probe lacks formal theoretical justification, and a sensitivity study on the choice of frequency range and perturbation strength is needed.

► [Online response \(AE1\)](#) | ► [PDF comparison](#)

► Response:

We now ground the Fourier probe in a local directional-sensitivity analysis and test its shortcut sensitivity directly with controlled experiments. PCS is the positive drop in the original predicted-class probability, and the expansion in Response R1.1 explicitly retains this positive-part operation. The analysis identifies what PCS measures—one-sided sensitivity along the structured Fourier directions—while the shape–texture interventions, paired shortcut counterfactuals, matched-coverage ImageNet-C purity analysis, and Colored-MNIST jointly establish that probe responsiveness provides discriminative, shortcut-sensitive routing information beyond confidence alone. Figure 12(c) further shows that this behavior is not tied to a narrowly tuned frequency range or perturbation strength.

Changes in the manuscript: Section III-C, p. 4; Section IV-C, Tables IV–VI, pp. 8–9, and Fig. 12(c), p. 12. See Response R1.1.

AE2: Alternatives and failure cases for geometry repair

The routed geometry repair module should be compared against simpler alternatives, and potential failure cases when the routed set is small or the prior estimate is noisy should be discussed.

► [Online response \(AE2\)](#) | ► [PDF comparison](#)

► Response:

We added the requested alternatives and tested how RGR behaves as the proportion of routed-away samples changes. Table VII compares discarding routed-away samples, feature moment matching, prototype contrastive alignment, fixed-uniform-prior OT, and the proposed dynamic-prior OT while keeping the remaining DCF components unchanged. We further added a routing-ratio stress test from 10% to 90% routed-away samples and clarified the prior EMA, uniform interpolation, centroid momentum, and empty-set safeguards. These tests show stable performance when enough trusted samples remain, and a clear decline when routed-away samples dominate the batch.

Changes in the manuscript: Section III-D, pp. 4–5; Section IV-C, Tables VII–VIII, p. 10; Algorithm 1, p. 6. See Response R1.2.

AE3: Computational cost and practical approximations

Computational overhead, including FLOPs and memory, should be reported in detail alongside a discussion of possible approximations for practical deployment.

► [Online response \(AE3\)](#) | ► [PDF comparison](#)

► Response:

Table XI now provides a complete resource comparison for DCF and representative baselines, including FLOPs, peak GPU memory, per-image latency, and accuracy under the same A100 evaluation setting. Full DCF achieves 43.48% long-horizon average accuracy with 24.6 GFLOPs, 14,690 MB peak memory, and 5.20 ms/image. DCF-Lite reduces the cost to 9.4 GFLOPs and 6,080 MB with 3.19 ms/image while retaining 42.06% accuracy, corresponding to reductions of 61.8% in FLOPs and 58.6% in peak memory for only a 1.42 pp accuracy decrease. Figure 12(d) and Table IX further quantify low-iteration Sinkhorn and curvature-proxy approximations, making the accuracy–efficiency trade-off explicit.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI and Fig. 12(d), p. 12. See Response R1.3.

AE4: Behavior of the layer retention gates

The per-layer behaviour of the curvature-aware retention mechanism needs further analysis, for example through a heatmap of retention gates over time and corruption types.

► [Online response \(AE4\)](#) | ► [PDF comparison](#)

► Response:

Figure 9 now shows the layer-wise retention gates throughout the 225-domain stream and across corruption types. The gates vary with both network depth and corruption type: layer4 remains highly permissive to candidate updates, while earlier and intermediate groups show stronger temporary source anchoring under specific corruptions. The temporal trajectories recover after transient suppression, demonstrating that CLR preserves adaptation capacity through dynamic retention. Table IX further compares control granularities and alternative curvature estimators.

Changes in the manuscript: Section III-E, pp. 5–6; Section IV-C, Fig. 9 and Table IX, p. 10. See Response R1.4.

AE5: Statistical evidence for cross-domain transfer

Additionally, statistical significance should be reported for the cross-domain results, where some improvements are marginal.

► [Online response \(AE5\)](#) | ► [PDF comparison](#)

► Response:

Figure 11 now reports mean $\pm$ standard deviation over five matched runs, and the accompanying “Cross-Domain Transfer” paragraph reports paired significance tests against the strongest aggregate baseline in each transfer setting. On ImageNet-C, DCF achieves a mean gain over No Adapt of 30.30 ± 0.39 pp versus 27.95 ± 0.05 pp for AEA, with $t(4) = 14.30$ and $p = 1.39 \times 10^{-4}$. On DomainNet-126, DCF achieves 61.40% mean accuracy versus 60.13% for DeYO, with $t(4) = 12.01$ and $p = 2.75 \times 10^{-4}$. These paired tests establish statistically significant aggregate differences across matched runs; individual transfer cells are reported with their run-to-run variability.

Changes in the manuscript: Section IV-C, “Cross-Domain Transfer,” and Fig. 11, p. 11. See Response R1.6.

Response to Reviewer 1

We appreciate your positive assessment of the motivation and long-horizon results. Your comments motivated additional analytical grounding, controlled shortcut experiments, mechanism-level robustness studies, and deployment-oriented evaluation. We address the six comments individually below.

Reviewer 1: Comment Overview and Navigation

Item	Comment Title	Direct Jump
R1.1	Fourier probe justification and sensitivity	Jump → p. ??
R1.2	Geometry repair alternatives and failure modes	Jump → p. ??
R1.3	Computational overhead and lightweight deployment	Jump → p. ??
R1.4	Layer retention patterns and alternative proxies	Jump → p. ??
R1.5	Additional literature on uncertainty-aware modeling	Jump → p. ??
R1.6	Variability, significance, and transfer beyond ImageNet	Jump → p. ??

R1.1: Fourier probe justification and sensitivity

The paper empirically shows that the Fourier perturbation exposes stress-inert predictions, but lacks a formal justification of why frequency-domain stress specifically targets shortcut cues (e.g., texture biases) rather than semantic features. A theoretical analysis—or at least a controlled synthetic experiment where shortcuts are known (e.g., color vs. shape)—would strengthen the claim. Currently, the probe design (random frequency and direction) appears heuristic; a sensitivity study on the choice of frequency range and perturbation strength λ would also help practitioners set these parameters without extensive tuning.

► [Online response \(R1.1\)](#) | ► [PDF comparison](#)

► Response:

Thank you for this suggestion. We added a local sensitivity analysis, controlled experiments with known shortcut factors, matched-coverage routing diagnostics, and a broader sensitivity study. Together, these additions provide both an analytical interpretation of PCS and direct evidence that its stress responsiveness contributes reliable routing information beyond predictive confidence alone.

Local squared-response interpretation of PCS

Section III-C defines PCS as the positive drop in the probability assigned to the original predicted class under the stressed view. For a fixed input and original predicted class, let δ be the actual perturbation after clamping and g_x the input gradient of that class probability. To make the positive-part operation in Eq. (5) explicit, the local squared-response expansion is

$$\mathbb{E}_q[s(x)^2] = \mathbb{E}_q\left[\left(-g_x^\top \delta\right)_+^2\right] + O(\mathbb{E}_q\|\delta\|_2^3).$$

Thus PCS measures one-sided probability sensitivity along the directions excited by the Fourier probe. Let $Q_q(x) = \mathbb{E}_q[\delta\delta^\top]$ denote the perturbation second-moment matrix. The untruncated quadratic response is $g_x^\top Q_q(x) g_x$; for the positive-part PCS, the leading term retains the truncation and is bounded above by this quadratic response. The relation uses the actual post-clamp perturbation second moment and accommodates asymmetric perturbations.

From a spectral and representational perspective, macroscopic semantic shape and object topology in natural images are predominantly supported by low-frequency phase coherence and macroscopic spatial boundaries, whereas texture shortcuts (e.g., periodic surface grain or repeating fabric weaves) exhibit localized, periodic high- and mid-frequency spectral energy. Because the Fourier basis probe Φ_{f_c, ω_c} perturbs specific radial frequency shells and orientations without altering the macroscopic spatial coordinates of the object, predictions that over-rely on brittle periodic texture cues suffer substantial destruction of their critical frequency patterns (yielding large probability drops and high PCS). Conversely, predictions grounded in low-frequency shape contours remain robust to these phase-direction stresses, maintaining consistent classifications (as evidenced by the $\geq 99.4\%$ shape decision agreement in Table IV).

Under a locally smooth predicted-class probability and sufficiently small perturbations, this expansion characterizes the directional sensitivity measured by PCS. The controlled interventions establish the empirical bridge from directional sensitivity

to shortcut-aware routing: in the shape–texture benchmark, PCS separates task-relevant from shortcut-driven predictions among confidence-matched candidates, while the matched-coverage ImageNet-C analysis shows that adding PCS improves trusted-set purity beyond entropy alone; Colored-MNIST further confirms the value of the resulting routing strategy under a distinct controlled shortcut. Together, these results establish PCS as a structured, shortcut-sensitive stress-response signal that materially strengthens routing beyond predictive confidence alone. Its role in DCF does not require identifying every possible shortcut in isolation; it is sufficient that the structured response provides complementary evidence for deciding which confident predictions should drive adaptation.

Controlled shortcut evidence

We added a 224×224 synthetic benchmark in which circle/square shape is the task factor and four sinusoidal textures provide explicitly controlled shortcuts. Table IV reports both shape-decision agreement and separate shape/texture PCS responses, allowing us to measure differential probe sensitivity while verifying that the semantic shape decision is preserved.

Table IV. Factorized response diagnostics of the Fourier probe. Shape agreement measures consistency between original and stressed shape predictions, while PCS measures the response of the corresponding factor-specific readout. The shape/texture PCS ratio characterizes the probe’s differential sensitivity across visual factors. Lower, middle, and upper nominal bands correspond to $[1, 75)$, $[75, 150)$, and $[150, 224]$ cycles/image, respectively.

Probe	Shape agreement (%)	Shape PCS ¹	Texture PCS ¹	PCS ratio
Lower nominal band	99.40	241.0	69.0	$3.49\times$
Middle nominal band	99.60	224.0	62.0	$3.61\times$
Upper nominal band	99.70	217.0	58.0	$3.74\times$
Energy-matched pixel noise	99.80	112.0	98.0	$1.14\times$
Random Fourier probe	99.60	229.0	64.0	$3.58\times$

¹ PCS entries are in units of 10^{-3} . The three nominal bands are $[1, 75)$, $[75, 150)$, and $[150, 224]$ cycles/image. “Nominal” denotes the frequency sampling parameter before image-grid discretization.

Across preserved, randomized, and reversed shortcut correlations, PCS achieves 0.81 AUROC for distinguishing task-relevant from shortcut-driven predictions, compared with 0.71 for energy-matched pixel noise. At matched trusted-set coverage, shortcut contamination decreases from 24.6% to 11.8%. On ImageNet-C, Table VI further compares filters at the same frozen model state and matched coverage: pseudo-label purity is 49.9% for entropy + PCS, versus 44.0% for entropy alone and 47.1% for PCS alone. These results show that PCS is not merely a second threshold correlated with entropy: it captures shortcut-sensitive differences that remain informative after confidence matching and materially improves trusted-set quality at matched coverage. This is the specific sense in which the Fourier probe strengthens routing beyond confidence alone.

We also added binary Colored-MNIST as a distinct color-shortcut setting. Digit identity defines the semantic target; training labels contain 25% noise, while test labels are clean and color–label agreement is approximately 30%. DCF obtains 88.91% accuracy, compared with 62.37% for DeYO and 50.93% for the source model.

Summary of evidence from revised Tables V–VI and the controlled-shortcut analysis

Diagnostic	Comparator / reference	DCF / joint-routing result
Colored-MNIST accuracy (Table V)	Best listed baseline: DeYO 62.37%	88.91% (+26.54 pp)
Pseudo-label purity at matched coverage (Table VI)	Entropy only 44.0%; PCS only 47.1%	Entropy + PCS 49.9%
Shortcut contamination at matched coverage	Confidence-matched candidates: 24.6%	Trusted-set contamination: 11.8%

Frequency range and perturbation strength

Figure 12(c) jointly varies the nominal frequency sampling range and perturbation strength λ . Section III-C specifies independent per-channel frequency and direction sampling, RMS-normalized Fourier bases, and exponential amplitude sampling with rate $1/\lambda$; the default is $\lambda = 0.2$ with nominal frequencies sampled from $[1, 224]$ cycles/image. Performance remains stable across a broad neighborhood of this configuration, supporting the reported setting as a practical default requiring only coarse adjustment.

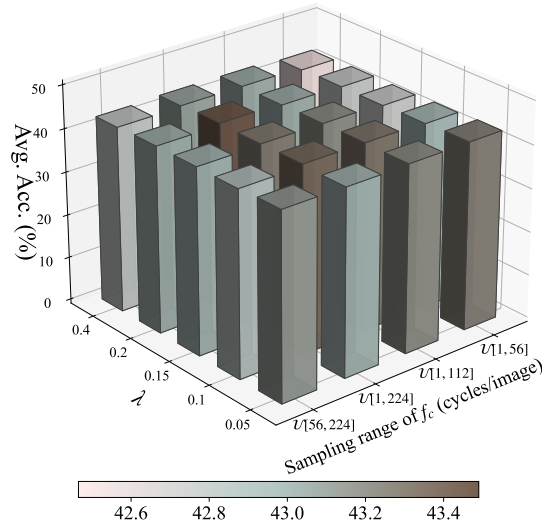


Fig. 12(c). Hyperparameter sensitivity. Sensitivity to perturbation strength λ and nominal frequency sampling range f_c (cycles/image).

Changes in the manuscript: Section III-C, p. 4, Eqs. (4)–(7); Section IV-C, “Analysis of Probe-Supported Sample Routing,” pp. 7–9; Tables IV–VI, pp. 8–9; Fig. 12(c), p. 12.

R1.2: Geometry repair alternatives and failure modes

RGR uses entropic OT with dynamic class priors to align routed-away features to source-initialized centroids. However, the paper does not compare this with simpler alternatives like feature moment matching or contrastive learning. Moreover, the centroid update (Eq. 19) uses only routed-away samples; if the routed set dominates, centroids may drift away from source semantics. The authors should discuss failure cases when the routed set is small or when the prior estimate is noisy (e.g., early adaptation stages). Adding an ablation with fixed uniform priors would isolate the benefit of dynamic priors.

► [Online response \(R1.2\)](#) | ► [PDF comparison](#)

► Response:

Thank you for suggesting these comparisons. We added them to separate the effects of OT-based geometry repair and the dynamic prior, and used a routing-ratio stress test to examine when RGR remains effective.

Comparison with simpler alternatives

Table VII keeps PSR and CLR fixed and changes only the treatment of the routed-away subset. It compares discarding those samples, feature moment matching, prototype contrastive alignment, OT with a fixed uniform prior, and OT with the proposed dynamic prior.

Table VII. Comparison of alternative treatments for routed-away samples on ImageNet-C. All variants share the same PSR and CLR.

Treatment of routed-away samples	T-CS (%)	T-CS-LS (%)	Long-horizon final (%)
Discard samples	41.87	42.71	42.88
Moment matching	42.81	42.82	41.51
Prototype contrastive alignment	42.83	42.79	42.33
OT with fixed uniform prior	43.01	42.80	42.91
OT with dynamic prior	43.49	43.29	43.84

The long-horizon final accuracy is 43.84% for dynamic-prior OT, compared with 42.91% for uniform-prior OT, a 0.93 pp improvement. Moment matching and contrastive alignment obtain 41.51% and 42.33%, respectively, versus 42.88% when routed-away samples are discarded. Thus, OT-based repair provides the strongest long-horizon result among the evaluated treatments, and the dynamic marginal contributes a further 0.93 pp gain over uniform-prior OT.

Conservative prior and centroid updates

Section III-D now spells out the prior construction. The batch prior uses stop-gradient predictions from the full batch, and its EMA starts from the uniform distribution with update rate $\rho = 0.01$. The OT class marginal then mixes this estimate with the uniform prior using $\alpha_b = 0.01$. Consequently, its L1 deviation from uniform is bounded by 0.02 and every class retains marginal mass of at least $0.99/K$, tightly regularizing transport allocation while still allowing target evidence to influence the class marginal.

The centroids are initialized from normalized rows of the frozen source classifier and updated outside backpropagation using soft OT-weighted feature means with momentum $\rho_M = 0.99$ (Eq. (16)), followed by normalization. If the numerical transport mass of a class is negligible, its previous centroid is retained. Source initialization and slow momentum anchor early geometry while allowing gradual target adaptation. Persistent target bias can still induce drift, motivating the routing-ratio stress test below.

Small and dominant routed-away subsets

The revised method also handles empty subsets explicitly. When the routed-away subset is empty, the geometry loss is zero and the centroids remain unchanged. When the trusted subset is empty, the entire adaptation step is skipped and the model, centroids, and prior state are retained. The routing-ratio stress test examines how performance changes as the balance between trusted supervision and routed-away geometry information shifts.

Table VIII. Routing-ratio stress test on ImageNet-C (T-CS). Controlled ratios are imposed per mini-batch using PSR-based ranking; Native uses the original routing. Acc. denotes average top-1 accuracy, and centroid drift denotes the mean ℓ_2 prototype displacement from the initial online reference.

Routed-away proportion	T-CS accuracy (%)	Centroid drift
Native routing	43.49	0.0518
10%	42.98	0.0052
25%	43.49	0.0217
50%	43.18	0.0829
75%	41.64	0.1268
90%	22.12	0.0468

The controlled 10%–50% routed-away regime yields 42.98%–43.49% accuracy. Performance decreases to 41.64% at 75% and to 22.12% at 90%, showing that a dominant routed-away subset leaves insufficient trusted evidence for consistency learning. Centroid drift is smaller at 90% than at 75% despite the sharper accuracy loss, identifying trusted-supervision scarcity as a distinct failure mechanism. Together, these results identify a broad stable routing regime and show why trusted consistency supervision and routed-away geometry repair must operate together.

Changes in the manuscript: Section III-D, pp. 4–5, Eqs. (10)–(16); Algorithm 1, p. 6; Section IV-C, “Analysis of Routed-away Geometry Repair,” pp. 9–10; Tables VII–VIII, p. 10. The centroid update referred to as Eq. (19) in the original review is Eq. (16) in the revision.

R1.3: Computational overhead and lightweight deployment

DCF involves Sinkhorn iterations for OT, curvature estimation per layer, and multiple forward passes for stressed views, leading to higher training memory and runtime (Fig. 12 shows $\sim 0.005\text{s/image}$ on A100). For edge deployment, this may be prohibitive. The paper should report FLOPs or memory usage in detail and discuss possible approximations (e.g., reducing Sinkhorn iterations, using cheaper curvature proxies) without sacrificing stability. A comparison with lightweight baselines (e.g., BN Adapt) in terms of throughput would contextualize the trade-off.

► [Online response \(R1.3\)](#) | ► [PDF comparison](#)

► **Response:**

We agree that runtime alone is insufficient to assess deployment cost. We therefore replaced the original accuracy–runtime comparison with a full resource profile and introduced a measured lightweight configuration, DCF-Lite.

All measurements use an NVIDIA A100, ResNet-50, and ImageNet-C under the long-horizon T-CS protocol, with batch size 64. The comparison includes BN Adapt and representative lightweight or stable baselines so that the accuracy–resource trade-off can be evaluated under a consistent setup.

Table XI. Computational overhead and lightweight variant. Comparison of FLOPs, peak memory, per-image runtime, and average accuracy on ImageNet-C (long-horizon T-CS, ResNet-50) measured on an NVIDIA A100.

Method	FLOPs (G)	Peak memory (MB)	Latency (ms/image)	Reported accuracy (%)
No Adapt	4.1	1,834	0.50	7.19
BN Adapt	4.1	1,944	0.50	29.83
Tent	8.2	5,685	1.40	15.79
CoTTA	24.6	13,892	2.80	34.54
SAR	16.4	6,936	3.00	38.90
DeYO	10.0	5,895	2.10	24.18
AEA	8.3	6,936	1.60	7.60
PTTA	16.4	11,872	4.10	22.63
RoTTA	28.9	12,952	18.70	16.39
LAW	16.4	13,364	6.90	26.54
TRIBE	65.6	18,405	15.60	25.07
SPA	49.1	9,635	10.30	12.50
DCF full	24.6	14,690	5.20	43.48
DCF-Lite	9.4	6,080	3.19	42.06

Relative to BN Adapt, full DCF improves the reported accuracy by 13.65 pp while incurring additional computation. For throughput context, the reciprocal of the reported per-image latency gives approximate rates of 2,000 images/s for BN Adapt, 192 images/s for full DCF, and 313 images/s for DCF-Lite.

Measured approximations

The revised manuscript describes four cost-saving choices for DCF-Lite: removing persistent source/candidate model copies and per-sample gradients, retaining gradients only through the clean adaptation branch, using a detached Fourier probe for PSR routing and transport assignment, and reducing N_{sk} to 1. Lite preserves the source parameter anchor as a tensor snapshot and caches a fixed curvature reference before the first optimizer update. This reference is the element-wise squared gradient of the first adaptation loss, capped element-wise at 10^{-4} , and is reused in subsequent comparisons with the current curvature proxy for retention-gate computation.

On the same A100 setup, Lite reduces FLOPs from 24.6 G to 9.4 G, peak memory from 14,690 MB to 6,080 MB, and latency from 5.20 to 3.19 ms/image, achieving 42.06% accuracy—1.42 pp below full DCF and 3.16 pp above SAR. Figure 12(d) supports low Sinkhorn iteration counts across the tested regularization range. Table IX shows only 0.09/0.22 pp gains from Expected Fisher/Hutchinson proxies at an additional 1.9/6.8 ms/image, favoring the default proxy’s accuracy–cost trade-off.

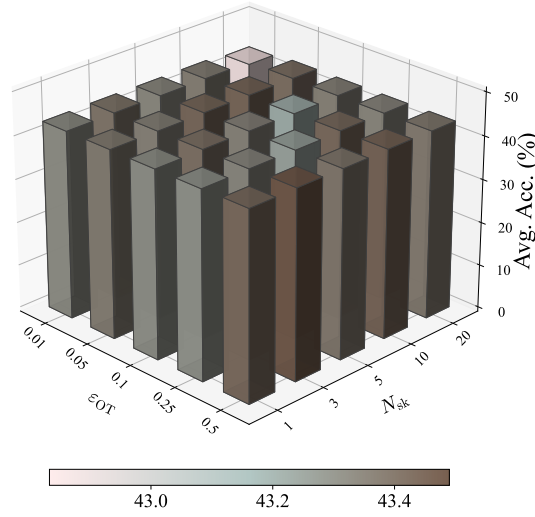


Fig. 12(d). Hyperparameter sensitivity. Stability across Sinkhorn iterations N_{sk} and entropic regularization ϵ_{OT} , confirming reliable OT alignment with low iteration counts.

These choices retain the route–adapt–retain design and reduce FLOPs by 61.8%, peak memory by 58.6%, and per-image latency by 38.7%. The resulting 42.06% accuracy, only 1.42 pp below full DCF, further shows that the long-horizon performance benefit of the route–adapt–retain principle is not tied to the full-cost implementation and can be largely preserved under substantially cheaper approximations.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI, p. 12; Table IX, p. 10; Fig. 12(d), p. 12. Table XI supersedes the runtime-only comparison cited as Fig. 12 in the original review.

R1.4: Layer retention patterns and alternative proxies

CLR uses curvature mismatch to gate retention, but the paper does not analyze which layers (early vs. late) are typically retained or pulled back. Figure 8 shows drift trajectories but not per-layer retention gates μ_t^l . Providing a heatmap of μ_t^l over time and across corruption types would reveal whether the method consistently preserves low-level (e.g., first conv) or high-level (classifier) layers. Additionally, the curvature proxy is estimated from pseudo-label gradients, which themselves may be noisy; evaluating CLR with Fisher information or Hessian approximations would be insightful.

► [Online response \(R1.4\)](#) | ► [PDF comparison](#)

► Response:

We added the requested gate visualization and comparisons with alternative curvature estimators. The revised analysis separates parameter drift, shown in Fig. 8, from the retention gate itself, shown directly in the new Fig. 9.

Which groups are retained and when

CLR acts on the trainable normalization affine parameters. For ResNet, these parameters are grouped into the stem normalization group (“initial_bn”) and layer1–layer4; for ViT, grouping follows encoder blocks.

Figure 9(a) follows the gates over the long-horizon stream, and Fig. 9(b) summarizes their corruption-dependent behavior. Layer4 remains near 0.98, while the stem and layer1 generally lie around 0.93–0.96, layer2 around 0.95–0.96, and layer3 can fall to approximately 0.92 under stronger mismatch. Motion most strongly suppresses the stem gate, Glass affects layer1, and Frost produces the strongest reduction in layer3. The transient suppression around domain indices 30–45 subsequently recovers toward 0.95–0.96, showing that adaptive source anchoring preserves plasticity after transient mismatch.

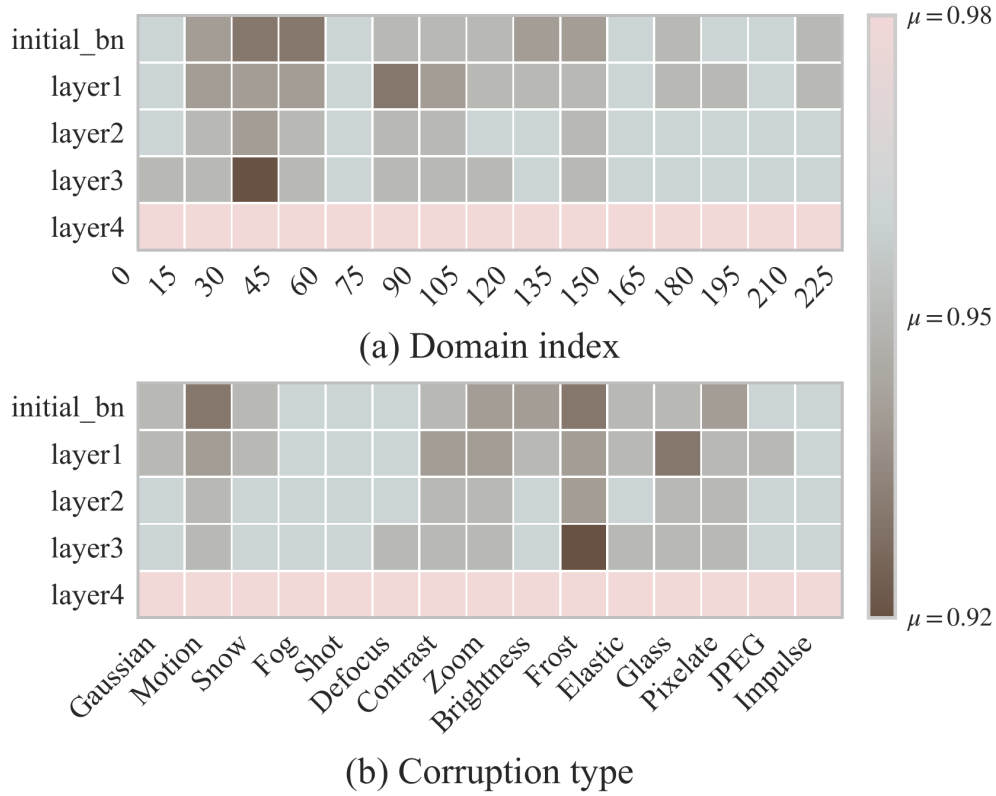


Fig. 9. Per-layer retention gates μ_t^l of CLR (ResNet-50). (a) Gates over the long-horizon stream (Domain index $0 \rightarrow 225$). (b) Gates per corruption type on T-CS.

A larger gate favors the candidate update and a smaller gate strengthens source anchoring. In the barycentric solution in Eq. (18), the gate and proxy vectors jointly determine the effective retention weights. The observed pattern therefore reflects the measured source–candidate mismatch within each layer group.

Granularity and curvature proxy comparisons

Table IX. Ablation of design choices in CLR on ImageNet-C (long-horizon T-CS). We compare different control granularities and curvature-aware proxy estimators.

Design choice	Reported accuracy (%)	Extra latency (ms/image)
Per-normalization-layer control	42.92	0
Per-layer-block control (default)	43.48	0
Per-sample gradient-square proxy (default)	43.48	0
Expected Fisher proxy	43.57	1.9
Hutchinson-based diagonal approximation	43.70	6.8

The group comparison favors layer-block control by 0.56 pp. Expected Fisher and the Hutchinson-based approximation reach 43.57% and 43.70%, compared with 43.48% for the default gradient-square proxy, at an additional 1.9 and 6.8 ms/image. The small accuracy spread across substantially more expensive estimators indicates that CLR’s benefit is not dependent on a particular high-cost curvature approximation. The default gradient-square proxy therefore provides the strongest accuracy–cost balance among the evaluated choices, giving up only 0.09/0.22 pp relative to Expected Fisher/Hutchinson while avoiding their additional 1.9/6.8 ms per image.

For full DCF, Section III-E clarifies that the candidate and source proxies are computed on the same current unlabeled batch at their respective parameter states. For each layer group, Eq. (19) averages the element-wise squared per-sample gradients over the batch, producing a diagonal proxy vector; Eq. (20) then normalizes the squared source–candidate proxy mismatch by the group’s parameter dimension. DCF-Lite uses the fixed reference cached before the first optimizer update, as described in Response R1.3, and compares it with the current curvature proxy to compute retention gates.

Changes in the manuscript: Section III-E, pp. 5–6, Eqs. (17)–(20); Section IV-A, “Models and Implementation Details,” p. 6; Section IV-C, “Analysis of Curvature-aware Layer Retention,” Fig. 9 and Table IX, p. 10.

R1.5: Additional literature on uncertainty-aware modeling

The literature review part can be improved by including more recent publications on machine learning methods, which can refer to Uncertainty-aware, high-precision multi-step prediction of structural health monitoring sensor streams under extreme typhoon events: an enhanced Bayesian dynamic linear model leveraging the kernel regression basis function for severe environmental adaptation, Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review, Towards high-accuracy data modelling, uncertainty quantification and correlation analysis for SHM measurements during typhoon events using an improved most likely heteroscedastic Gaussian process.

► [Online response \(R1.5\)](#) | ► [PDF comparison](#)

► Response:

Thank you for suggesting these studies. We have incorporated all three references into Section II as cross-disciplinary work on uncertainty-aware modeling under evolving environmental conditions. The added references are:

[19] Q.-A. Wang et al., “Uncertainty-Aware, High-Precision Multi-Step Prediction of Structural Health Monitoring Sensor Streams under Extreme Typhoon Events: An Enhanced Bayesian Dynamic Linear Model Leveraging the Kernel Regression Basis Function for Severe Environmental Adaptation,” *Measurement*, vol. 262, article 120050, 2026.

[20] Q.-A. Wang, A.-W. Lu, Y.-Q. Ni, J.-F. Wang, and Z.-G. Ma, “Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review,” *Sensors*, vol. 25, no. 12, article 3577, 2025.

[21] Q.-A. Wang et al., “Towards High-Accuracy Data Modelling, Uncertainty Quantification and Correlation Analysis for SHM Measurements during Typhoon Events Using an Improved Most Likely Heteroscedastic Gaussian Process,” *Smart Structures and Systems*, vol. 32, no. 4, pp. 267–279, 2023.

These studies contribute Bayesian dynamic modeling, Bayesian-network reasoning, and heteroscedastic uncertainty quantification for non-stationary structural-health-monitoring streams. We added them to broaden the discussion of uncertainty under evolving environments. DCF addresses the corresponding online adaptation problem in visual recognition, where the model itself is updated from unlabeled, temporally correlated target observations; its contribution is the joint control of target evidence, routed-away geometry, and layer-wise update retention.

Changes in the manuscript: Section II, “Adaptation without Target Data,” p. 3; references [19]–[21], p. 13.

R1.6: Variability, significance, and transfer beyond ImageNet

In Figure 10, DCF shows positive gains across all source domains, but the paper does not report the variance or statistical significance. Some cells show marginal improvements (<1%), which may not be robust. The authors should include standard deviations over multiple runs and perform a significance test (e.g., paired t-test) against the best baseline. Furthermore, the cross-domain evaluation adapts on one corruption and tests on the remaining 14, but the test set still shares the same ImageNet classes; evaluating on a completely different dataset (e.g., CIFAR-10-C or a medical imaging shift) would better demonstrate generalizable representation preservation. Adding such an experiment or discussing the limitation would strengthen the paper’s claims.

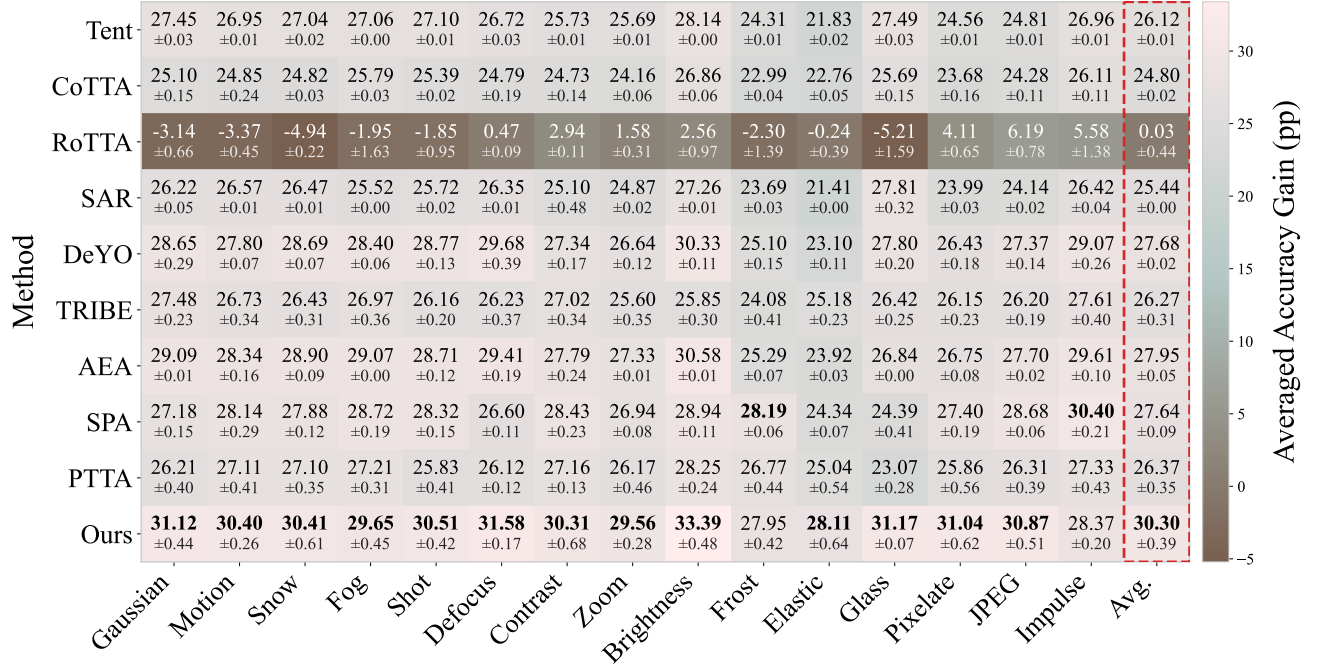
► [Online response \(R1.6\)](#) | ► [PDF comparison](#)

► Response:

We agree that the cross-domain results should be supported by run-to-run variability and formal significance testing. We therefore added five matched runs, paired significance tests, and a second transfer benchmark on DomainNet-126. The resulting analysis shows that DCF’s aggregate transfer advantage is statistically significant on both ImageNet-C and DomainNet-126, while the per-transfer mean $\pm$ standard deviation values characterize how those gains are distributed across individual transfer settings.

Variability across five runs and paired comparisons

The revised Fig. 11 replaces the original Fig. 10. Panel (a) updates the existing ImageNet-C analysis with five-run variability, while panel (b) adds the new DomainNet-126 benchmark. Both panels report mean $\pm$ standard deviation over five matched runs. On ImageNet-C, DCF achieves a mean gain over No Adapt of 30.30 ± 0.39 pp, compared with 27.95 ± 0.05 pp for AEA, the strongest aggregate baseline in this comparison.



Adaptation Domain (Evaluated on the Remaining 14 Unseen Corruptions)

Fig. 11(a). **Cross-domain transfer.** ImageNet-C accuracy gains (in percentage points) over No Adapt, averaged across all unseen corruption domains. Results are reported as mean $\pm$ standard deviation over five runs.

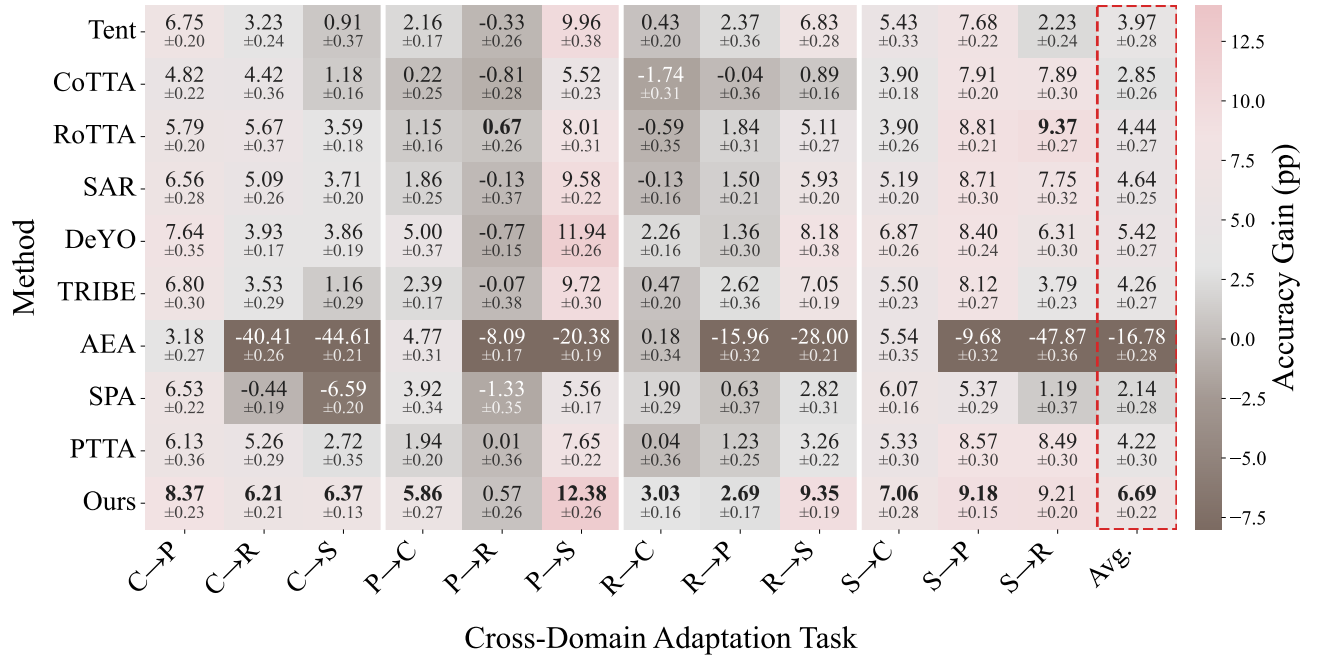


Fig. 11(b). **Cross-domain transfer.** DomainNet-126 accuracy gains over No Adapt for each cross-domain transfer pair (Adaptation $\rightarrow$ Evaluation, where C: Clipart, P: Painting, R: Real, and S: Sketch). Results are reported as mean $\pm$ standard deviation over five runs.

Summary of cross-domain statistical results reported in revised Section IV-C

Evaluation	DCF	Best aggregate baseline	Difference	Paired test
ImageNet-C gain over No Adapt	30.30 ± 0.39 pp	AEA / 27.95 ± 0.05 pp	+2.35 pp	$t(4) = 14.30$; $p = 1.39 \times 10^{-4}$
DomainNet-126 mean accuracy	61.40%	DeYO 60.13%	+1.27 pp	$t(4) = 12.01$; $p = 2.75 \times 10^{-4}$

The paired tests establish statistically significant improvements in the matched run-level aggregates on both benchmarks. The ImageNet-C test compares mean gains over No Adapt, while the DomainNet-126 test compares mean accuracies. Statistical inference is intentionally conducted at this aggregate run level, which matches the reported cross-domain summary metric; the per-transfer cells are reported with mean $\pm$ standard deviation to transparently characterize the distribution of gains rather than being overinterpreted as independently powered significance tests.

Fig. 11(a): Domain-aggregated transfer on ImageNet-C. Each entry reports transfer gain after adaptation to one of the 15 corruption domains, averaged across the remaining 14 unseen corruptions without further updates. DCF exceeds the domain-wise strongest baseline in 13 of 15 domains, with advantages above 1.5 pp in 12 domains and a maximum of 3.64 pp. The small-gain or lower-performing cases are Fog (+0.58 pp over AEA), Frost (-0.24 pp versus SPA), and Impulse (-2.03 pp versus SPA). These domain-wise results describe the distribution of gains; statistical significance is evaluated by the five matched run-level aggregate comparisons reported above.

Fig. 11(b): Pairwise transfer on DomainNet-126. Each entry reports mean $\pm$ standard deviation over five runs for a specific Adaptation $\rightarrow$ Evaluation pair. DCF exceeds DeYO in all 12 pairs and yields positive gains over No Adapt throughout this matrix (0.57–12.38 pp). It achieves the highest mean gain in 10 of 12 pairs; the exceptions are P $\rightarrow$ R (0.57 versus RoTTA’s 0.67 pp) and S $\rightarrow$ R (9.21 versus RoTTA’s 9.37 pp). This consistent positive transfer, together with the significant aggregate advantage over DeYO, provides evidence that DCF preserves transferable representations beyond the ImageNet corruption setting and across the evaluated artistic-domain shifts.

Transfer on a distinct dataset

To extend the evaluation beyond ImageNet corruptions, we added DomainNet-126, a separate 126-class benchmark spanning Clipart, Painting, Real, and Sketch. Models adapt on one domain and are then evaluated on the remaining unseen domains without further updates. Each benchmark uses its own fixed label space and source classifier. This experiment therefore tests whether the representation-preservation behavior observed on ImageNet-C carries to a distinct dataset and a different family of domain shifts, rather than only to held-out corruptions within the same benchmark.

Changes in the manuscript: Section IV-A, “Benchmarks,” p. 6; Section IV-C, “Cross-Domain Transfer,” and Fig. 11(a)–(b), p. 11.

Response to Reviewer 2

Thank you for the positive assessment of the route–adapt–retain framework and the long-horizon evidence. We have addressed the reproducibility, visualization, practical-guidance, and literature suggestions as follows.

Reviewer 2: Comment Overview and Navigation

Item	Comment Title	Key Revision	Direct Jump
R2.1	Trainable parameters and the meaning of a layer	Tent affine setup; ResNet/ViT block grouping	Jump → p. ??
R2.2	Explicit definitions of the four routing regions	Mathematical inequalities for Areas 1–4	Jump → p. ??
R2.3	Practical threshold guidance and OT settings	Default threshold guidance and OT specifications	Jump → p. ??
R2.4	Recent work on continual adaptation	Integrated GOLD and recent continual-TTA survey	Jump → p. ??

R2.1: Trainable parameters and the meaning of a layer

The implementation section states that the experiments follow the official Tent setup. For clarity and reproducibility, please explicitly specify which model parameters are optimized during TTA (e.g., all network parameters, only the affine parameters of normalization layers, or another subset). It would also be helpful to clarify how a layer is defined when applying Curvature-aware Layer Retention. For convolutional architectures, does this refer to an individual normalization layer, a residual block, or a larger network stage?

[► Online response \(R2.1\)](#) | [► PDF comparison](#)

► Response:

We now specify which parameters are optimized and how they are grouped for CLR. Following Tent, SGD updates only the affine parameters of normalization layers, with learning rate 2.5×10^{-4} and momentum 0.9; convolution kernels and classifier weights remain frozen during adaptation.

For ResNet, CLR groups these trainable affine parameters into the stem normalization group (“initial_bn”) and the layer1–layer4 architectural groups. For ViT, grouping follows encoder blocks. This definition matches the layer labels used in the new gate visualization.

We additionally compare this block-level grouping against per-normalization-layer control. Table IX reports 43.48% for layer-block control and 42.92% for per-normalization-layer control, supporting the block-level grouping used by default.

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-A, p. 3; Fig. 9 and Table IX, p. 10.

R2.2: Explicit definitions of the four routing regions

The PCS–entropy visualization in Fig. 7 is a useful addition and provides good intuition for the routing mechanism. The caption already identifies Area 1 as the low-entropy/high-PCS trusted region, but the exact conditions defining all four regions are not stated explicitly. I suggest defining Areas 1–4 and their corresponding entropy/PCS threshold conditions directly in the figure caption or in the accompanying text. This would allow readers to interpret the visualization without repeatedly referring back to the routing equations.

[► Online response \(R2.2\)](#) | [► PDF comparison](#)

► **Response:**

We have explicitly defined all four regions in the sample-routing text in Section III-C, immediately after Eqs. (6)–(7). Let E denote predictive entropy, s denote PCS, and v_{Ent} and v_{PCS} denote the two thresholds. The four regions are defined by the corresponding entropy/PCS inequalities shown below.

Definitions added to revised Section III-C

Region	Entropy condition	PCS condition	Role
Area 1	$E < v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Trusted
Area 2	$E \geq v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Routed away
Area 3	$E < v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away
Area 4	$E \geq v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away

Only Area 1 supplies trusted consistency supervision, while Areas 2–4 form the routed-away set used for geometry repair. The explicit inequalities also highlight the role of Area 3: these samples appear confident according to entropy but exhibit insufficient stress responsiveness, precisely the regime in which entropy-only selection can retain stress-inert predictions.

Changes in the manuscript: Section III-C, p. 4, following Eqs. (6)–(7); Fig. 7, p. 9, provides the corresponding PCS–entropy diagnostic visualization.

R2.3: Practical threshold guidance and OT settings

The sensitivity analysis suggests that DCF remains effective across a reasonable range of routing hyperparameters. It would be useful to provide brief practical guidance, or a simple rule of thumb, for selecting the entropy and PCS thresholds when applying DCF to a new dataset. In addition, for reproducibility, please consider reporting the remaining implementation details of the optimal transport procedure, such as the number of Sinkhorn iterations and other relevant solver settings.

► **Online response (R2.3)** | ► **PDF comparison**► **Response:**

We now provide a starting rule for the routing thresholds and specify the OT solver settings. For a new K -class dataset, we recommend $v_{\text{PCS}} = 0.2$ and $v_{\text{Ent}} = 0.6 \ln K$ as the default starting point, followed by only coarse adjustment if necessary. The entropy threshold scales with the maximum classification entropy, $\ln K$, while the PCS threshold controls the required probability response to the Fourier stress probe. For the default OT solver, the revised manuscript specifies Sinkhorn–Knopp with $N_{\text{sk}} = 3$ iterations and entropic regularization $\varepsilon_{\text{OT}} = 0.5$; Figure 12(d) additionally evaluates lower iteration counts and alternative regularization values.

Figure 12(b) shows a broad high-accuracy region around this default setting, supporting it as a robust practical initialization. The routing-ratio analysis in Table VIII further explains the trade-off between trusted-set selectivity and the amount of reliable supervision available to the online update.

Implementation settings specified in the revised manuscript

OT or routing quantity	Value or specification
Sinkhorn–Knopp iterations	$N_{\text{sk}} = 3$
Entropic regularization	$\varepsilon_{\text{OT}} = 0.5$
Transport cost	1 - cosine similarity of normalized features and centroids
Sample marginal	Uniform over the routed-away subset
Class marginal	EMA prediction prior mixed with the uniform prior
Prior EMA and uniform mixing	$\rho = 0.01$; $\alpha_b = 0.01$; uniform initialization
Centroid momentum	$\rho_M = 0.99$, followed by normalization
Gradient treatment	Transport plans and prior predictions are detached
Routing defaults	$v_{\text{PCS}} = 0.2$; $v_{\text{Ent}} = 0.6 \ln K$

OT or routing quantity	Value or specification
Fourier amplitude default	$\lambda = 0.2$

Section III-D further specifies the shared centroids, two-view transport plans, cross-view geometry objective, empty-subset safeguards, and negligible-class-mass handling. Figure 12(d) jointly studies Sinkhorn iterations and entropic regularization. Stable downstream accuracy across the tested configurations supports efficient finite-iteration OT for online geometry repair. See Response R1.3 (p. ??) for Fig. 12(d).

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-D, pp. 4–5, Eqs. (10)–(16); “Hyperparameter Sensitivity,” p. 11, and Fig. 12(b), (d), p. 12.

R2.4: Recent work on continual adaptation

Given the rapid development of continual and long-horizon TTA, the related-work discussion could potentially be strengthened by briefly acknowledging very recent studies on long-horizon stability, where relevant. For example:
 [1] *The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation*, CVPR 2026.
 [2] *Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions*, TMLR 2026.

► [Online response \(R2.4\)](#) | ► [PDF comparison](#)

► Response:

Section II now cites both suggested 2026 studies and explains how their approaches relate to DCF.

[38] G. Lai, D.-W. Zhou, Z. Li, and H.-J. Ye, “The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation,” CVPR, 2026.

GOLD constrains updates to a classifier-aligned low-rank subspace. DCF takes a complementary approach by controlling adaptation evidence, reusing routed-away geometry, and selectively retaining candidate updates across layer groups according to source mismatch.

[28] S. K. Maharana et al., “Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions,” Transactions on Machine Learning Research, 2026.

The survey provides context for teacher models, memory, reset strategies, and long-horizon stability. DCF specifically links heterogeneous target evidence to layer-wise update retention.

Changes in the manuscript: Section II, “Test-Time adaptation” and “Reliability- and Stability-aware TTA,” p. 3; references [28] and [38], p. 13.